

CS & IT ENGINEERING



Compiler Design

Parsers

*finding Tokens
(first & follow)*

DPP – 01



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#Q. Consider the following C program:

```
int main ( ) { → 5  
    /*finding maximum element out of a & b*/ ○  
    int a, b, max; → 7  
    a = 10; b = 20; → 8 ✓  
    if (a < b) → 6 ✓  
        max = b; → 4 ✓  
    else  
        max = a; → 5 ✓  
    return (max); → 5 ✓  
} → 1 ✓
```

Calculate the total number of tokens present in the program?

41

#Q. Consider the following C-program:

```
1. int main( )  
2. {  
3. x = a + b * c;  
4. y = x + a;  
5. char f = 'e';  
6. int g = 200;  
7. ch/*comment ar = "gate";  
8. }
```

*/*comment*

→ lexical error

Which of the following is/are correct regarding above program?

Continue to Next Slide...

- A** The given program has 47 tokens. ~~X~~
- B** Given program produces compilation error
- C** Given program produces lexical error
- D** No error present in the program

[MCQ]



#Q. Compiler's first phase makes use of following patterns for token (S_1, S_2, S_3) recognition over the alphabet a, b, c.

$$\left. \begin{aligned} S_1: b\#(b|a)^*c &= (\epsilon + b) (\underline{b+a})^* c \\ S_2: c\#(c|b)^*a &= (\epsilon + c) (\underline{c+b})^* a \\ S_3: a\#(a|c)^*b &= (\underline{a+\epsilon}) (\underline{a+c})^* b \end{aligned} \right\}$$

abbbccccbba

abbbccccbba

Note: $x\#$ means 0 or 1 occurrence of the symbol x . The analyzer outputs the token that matched the longest possible prefix of the string. If abbbccccbba is processed by first phase of compiler then which one of the following is the sequence of token of output.

A

S_1, S_2, S_3

B

S_1, S_2

C

S_3, S_2 ✓

D

S_3, S_3

a b b b c c c c b a
 $S_3 S_2$

#Q. Consider the following C-program:

```

int main ( ) { → 5
    int x; /*comment*/ ⇒ 3
    x == y /*abcd*/; ⇒ 5
    int *p; → 4
    int b = 10, y; → 7
    x = *p +++++ y; → 9
} → 1

```

+ + + + +
+ + + + +

How many tokens are present in the given program?

34

#Q. Consider the following code:

```
main ()  
{  
    x = a/b + c;  
    y = c - a;  
    x = a + d;  
    y = d/c + b;  
}
```

Handwritten token counts in red:

- main () → 3
- { → 1
- x = a/b + c; → 8
- y = c - a; → 6
- x = a + d; → 6
- y = d/c + b; → 8
- } → 1

Summary of counts: 3 + 1 + 8 + 6 + 6 + 8 + 1 = 33

Calculate the total number of token in the above code.

33

#Q. Consider the following grammar G:

$$G: S \rightarrow [S \mid SS]$$

Where S is start variable, Number of terminals in follow set of S is 3.

$$S \rightarrow [S \mid SS]$$

$$\{ [, (,] \}$$

$$S \rightarrow a S / SS / b$$

$$\text{First}(S) = [,]$$

#Q. Consider the following grammar.

$$S \rightarrow (S) \mid \underline{SS} \mid \epsilon$$

Total number of terminals in follow set of S is 3.

$\{$

$$F(S) = \{), (, \$\}$$

$$\text{first}(S) = \{ (, \$$$

#Q. Consider the following Grammar G:

$S \rightarrow ACB$

$A \rightarrow \underline{a} \mid \underline{b} \mid \epsilon$

$B \rightarrow c \mid d \mid \epsilon$

$C \rightarrow \underline{e} \mid \underline{f}$

$$\begin{aligned} \text{first}(S) &= \text{first}(ACB) \\ &= \{a, b\} \text{ first}(C\underline{B}) \\ &\quad \{e, f\} \quad \{c, b, e, d\} \end{aligned}$$

How many terminals are present in First (S)?

A

6

B

3

C

7

D

4



#Q. Which of the following is/are correct regarding FIRST & FOLLOW of the given grammar.

$$E \rightarrow TE'$$

$$E' \rightarrow +TE' \mid \epsilon$$

$$T \rightarrow FT'$$

$$T' \rightarrow *FT' \mid \epsilon$$

$$F \rightarrow \text{id} \mid (E)$$

A FIRST (T) = {id, (}
FOLLOW(T') = {+, \$,)}

B FIRST (E) = {id, (}
FOLLOW(E') = {\$,)}

C FIRST (E) = {id, (}
FOLLOW(F) = {+, \$,), *}

D FIRST (T') = {*, $\hat{\epsilon}$ }
FOLLOW (E) = {\$, *,)}

#Q. Consider the following grammar G:

G:

$S \rightarrow AaAb \mid BbBa$

$A \rightarrow \epsilon$

$B \rightarrow \epsilon$

$\text{First}(AaAb) \cup \text{First}(BbBa)$

$\{a\} \cup \{b\} = \{a, b\}$

Number of terminals in First(S) is/are 2.

[MCQ]



#Q. Consider the following grammar:

$$S \rightarrow P(RQ \mid RbQ \mid Qa)$$

$$P \rightarrow da \mid QR$$

$$Q \rightarrow g \mid \epsilon$$

$$R \rightarrow h \mid \epsilon$$

$$\text{Follow}(R) = \{g, \$, b, h, \epsilon\}$$

The number of elements in Follow (R) and the follows set of R respectively is?

A 4, {\$, b, g, h}

B 3, {b, g, h}

C 5, {\$, a, b, g, h}

D 3, {\$, b, g}

13 problems /
First 8 follow
Tokens (10)

THANK - YOU

